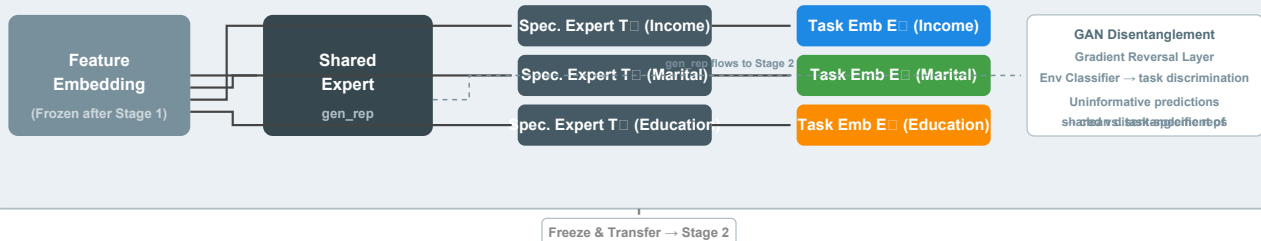


Stage 1: Multi-task Pre-training (all parameters frozen afterwards)

3-task backbone: $T_{\square} = \text{Income}$, $T_{\square} = \text{Marital}$, $T_{\square} = \text{Education}$ — 10 epochs, GAN env_coe = 0.1



Stage 2: New-task Prompt Tuning — Causal Manipulation (only projection network + tower trained, <10% params)

